Ziphozihle Mqopi

18/20

90%

Module #	671101000-KM-08:
NQF Level	level 4,
Notional hours	50
Credit(s)	Cr 5
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

LOW VOLTAGE PROTECTION

FORMATIVE ASSESSMENT

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About the Formative Assessment	

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.

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Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner:	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Learners Name: Ziphozihle Maqpi

Learner's signature: _____ Date: _____

Facilitator Name: _____

Facilitator's signature: _____ Date: _____

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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

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Item	Points to be covered	Tick
9	**Ensure the assessment environment is appropriate** or make special arrangements	
10	Discuss moderation	
11	Allow the learner opportunity to clarify any items discussed	

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: Ziphozihle Maqopi	Signature
Facilitator Name:	Signature:

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LEARNING UNIT 1:

KM-08-KT01: Low voltage protection (100%)

Learning Outcome 1:	KT0101 Purpose and application of low voltage protection
Learning Outcome 2:	KT0102 Types of low voltage protection
Learning Outcome 3:	KT0103 Low voltage protection parameters and statutory requirements

Task	Questions Description	Marks
1.	Describe the operation and functions of different types of low voltage protective devices including overload relays, fuses, circuit breakers and earth leakage protection devices.	10

1) Overload Relays

Operation: Overload relays are electronic devices that detect excessive current flow in a circuit. They typically operate on the principle of heat made by the current flowing through a bimetallic strip. When the current is too high the bimetallic strip bends.

Functions: Overload Protection: They protect electrical equipment from damage cause by excessive current

Motor Protection: They are used to protect motor from overheating and damage due to overload conditions

2) Fuses

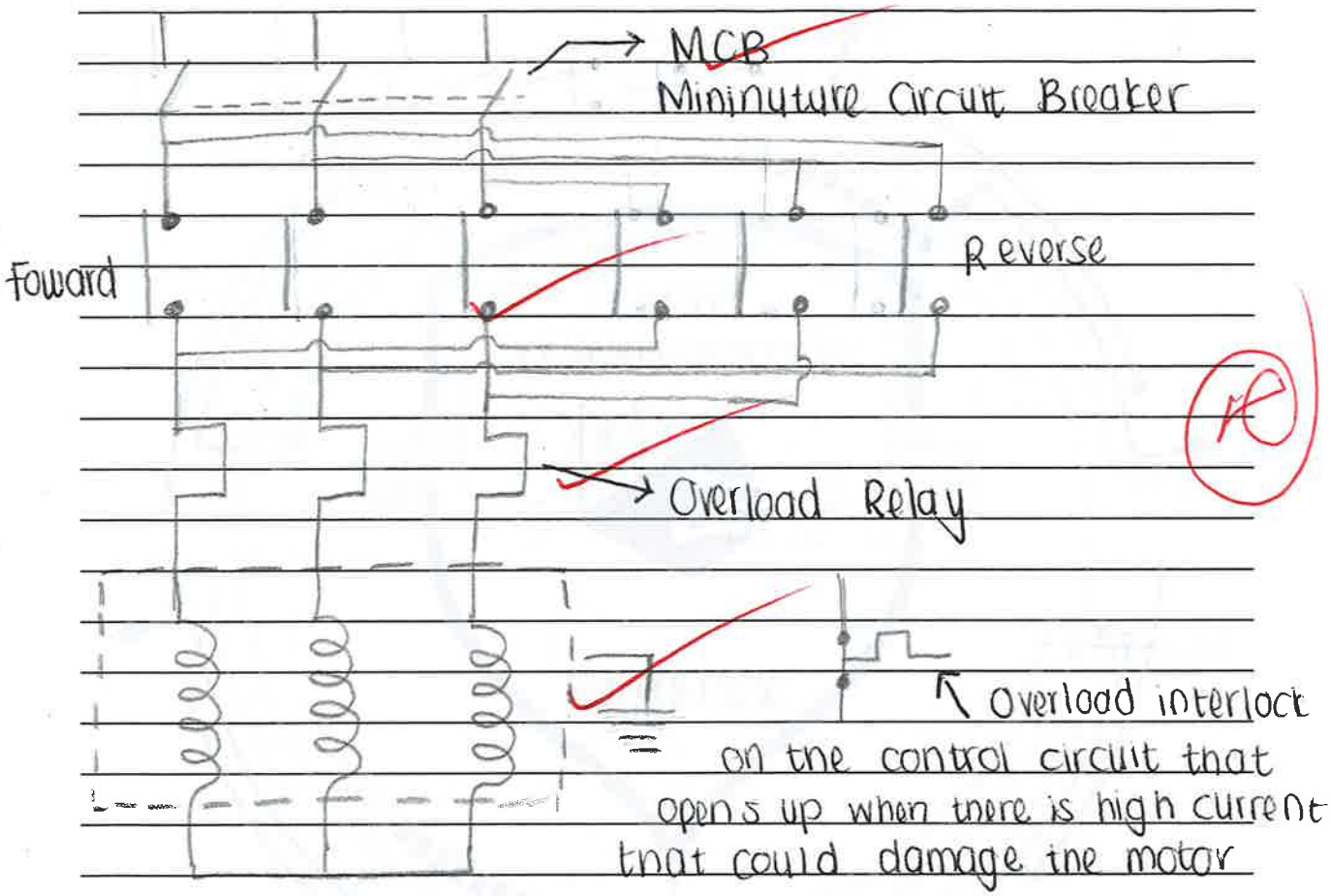
Operation: Fuses consits of a wire or strip of metal with a low melting point. When excessive current flows through the fuse the wire melts

Functions:

Overload Protection: They provide protection against overloads and short circuits preventing damage to electrical equipment

Task	Questions Description	Marks
2	Explain, with the aid of circuit diagrams, how single-and three phase electrical installations are protected.	10

Three Phase Foward Reverse Motor Connection



Miniature Circuit Breaker ~~(CBT~~ (MCB):
Uses a bimetalic strip. When current exceeds the rated value for too long the strip heats, bends and trips the latch

Overload Relay:
Thermal overload relay also uses a bimetallic strip or eutectic alloy. Prolonged currents above motor rating heats it, causing it to trip and open the contactor coil.

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SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

ASSESSMENT CRITERIA		1	2	3
1	KT0101 Purpose and application of low voltage protection			✓
2	KT0102 Types of low voltage protection			✓
3	KT0103 Low voltage protection parameters and statutory requirements			✓

Learner Signature **Date**

Facilitators' Signature **Date**

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DECLARATION BY LEARNER	
I, Ziphozihle Maopi .....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.	
Learner **Date**	**Facilitator** **Date**

DECLARATION BY THE ASSESSOR	
I ..... (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.	
Assessor **Date**	**Moderator** **Date**

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Notification of Assessment

Assessment will take place as follows:

	Theory	**Practical**
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

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